

Two stage Multi-Modal Modeling for Video Interaction Analysis in Deep Video Understanding Challenge

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ABSTRACT

Interaction understanding between different entities in human-centered movie video is receiving more and more attention. Recently, a deep video understanding (DVU) task is proposed to identify interactions between different person entities on scene level task. However, limited samples of DVU dataset and multiple complex interactions make it difficult. To tackle these problems, we propose a two stage multi-modal method to predict the interaction between entities. Specifically, scene segment is first divided into several sub-scene clips, meanwhile face trajectory and person trajectory are obtained through face tracking/recognition and skeleton-based person tracing. Then, we extract and jointly train multi-modal features in the same semantic space including face emotion feature, person entity feature, visual features, text features and audio features. Finally, zero-shot transfer model and multiple classification model are proposed to predict interactions together. The experimental results show that our method performs new state-of-the-art on the DVU dataset.

CCS CONCEPTS

• Computing methodologies → Scene understanding.

KEYWORDS

multi-modal feature, HLVU dataset, interaction prediction, two-stage method

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1 INTRODUCTION

With the increasing of videos uploaded online, automatic video analysis and understanding is an emergent and important task for human rapidly acquiring information from videos. A deep analysis of interactions between different entities in a long duration video

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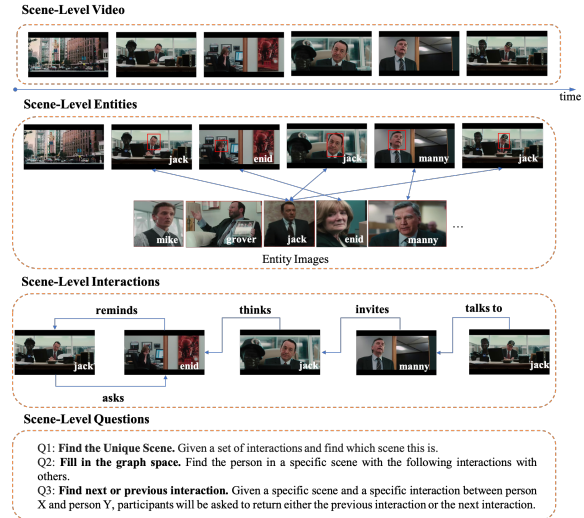


Figure 1: The definition of DVU task about interaction questions.

involves many video understanding technologies such as video scene segment [6, 23], object recognition [14], video visual relation detection [26], feature representation [22, 30] and so on.

Recently, a deep video understanding (DVU) task [7] is proposed which aims to construct a knowledge graph of a long duration video with given scene segments and types of the entities. Specially, DVU challenge contains movie-level task and scene-level task for evaluation. Meanwhile, a new HLVU dataset [7] is constructed, which contains 584 valid scene segments with corresponding annotations from 14 movie videos. In this paper, we focus on scene-level video analysis task, which aims to construct a knowledge of graph to predict interactions between different entities in a long scene segment.

According to the knowledge graph, three different types of questions can be answered as follows. (1) Finding the correct scene from multiple options with several interaction vocabularies; (2) Filling in the unknown character in the graph space given interactions between different entities; (3) Finding next or previous interaction in special scene from multiple options. The definition of DVU task about these questions is shown in Figure 1.

Interaction prediction between different entities in a scene-level clip has been explored recently by the community [4, 34]. However, all these tasks focus on only one interaction in a short clip on Kinetics [15] and ActivityNet [3] dataset but more interactions between different entities that appear in DVU task need to be predicted in a long duration scene segment. Moreover, limited samples of HLVU

dataset are difficult for training an effective model to predict interactions between different entities. Meanwhile, several interaction categories do not appear in the training set, it is also a challenge to classify unseen interaction categories in the testing set.

To address the above-mentioned difficulties in analysing long duration scene segments, we propose a two-stage method that utilizing multi-modal features of scene segment. For the first stage, we extract multi-modal features for each scene segment as the input for the next stage, including face emotion feature, skeleton-based person entities feature, visual feature, text feature and audio feature of scene segment. For the second stage, we design two models to predict all interactions between different entities in a scene segment with the input of multi-modal features respectively, and final prediction results are fused from two models.

The main contributions of this work are summarized as follows.

- With the problem of limited samples and missing data of HLVU dataset, we design a zero-shot transfer learning method together with multiple classification method to improve accuracy of interaction predictions.
- Multi-modal features are integrated in our two stage method, and its consistences in a semantic space are also guaranteed.
- With the two stage multi-modal method, we achieve new state-of-the-art results on HLVU dataset of DVU task.

2 RELATED WORKS

Face Recognition. Deep learning based face recognition has achieved significant progress in recent years. The processing of face recognition usually consists of multiple steps, such as face detection, key-points landmark detection, face alignment, feature extraction, and face recognition. A public groundwork is welcomed for deploying the face recognition in the shape of holistic pipeline. A novel open-source framework, named FaceX-Zoo [32], which contains a face SDK to easily upgrade and extend along with the development of face related domains.

Skeleton-Based Person Tracing. In human-centered videos, human-to-human interaction is often accompanied by interactions in body movements. The skeleton-based key points of person can characterize the posture and the action of the human body, and provide assistance for the tracking of the human body and the classification of human interaction. Faster R-CNN [24] is a commonly used framework for human body detection. Through the special design of the network structure, HRNet [33] can maintain the high-resolution characterization of the image, and achieves significant high accuracy for downstream attitude estimation and key point extraction. For person identification by tracking, pose-based naive tracker [11] correlate key-points of two persons based on inter-frame pose similarities.

Multimodal feature representation. As scene-level clip in DVU task has multiple sources such as video modal, audio modal, text modal from automatic speech recognition (ASR), features of scene can be represented by multi-modal features. Early visual features of videos usually extracted by convolution networks [4, 30]. With the advent of the transformer, videos are often processed by visual transformer [10] to get more robust features. After transforming speech in videos to text by ASR, text encoder such as Transformer [31] is often used to extract text feature of sentences.

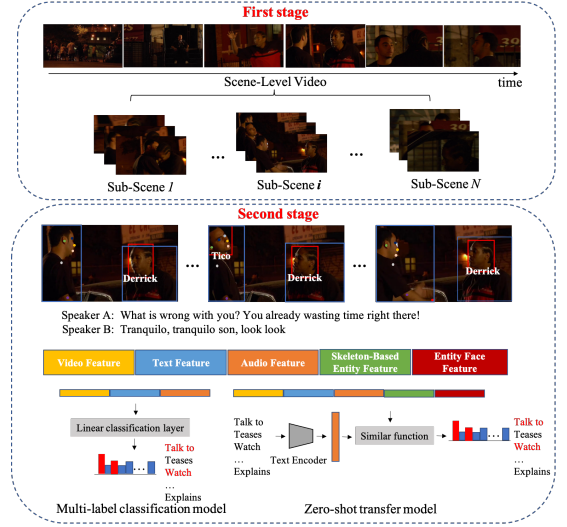


Figure 2: The structure of our method.

Clip-based Video-Language Pre-training and Finetune. Vision-Language understanding attracts increasing attention in the research community [17, 19, 22, 35]. Inspired by the rapid progress of large-scale image-text pre-training (e.g., CLIP [22]), we believe the knowledge learned from image-text pairs would be helpful for video understanding tasks. Several pioneering works have been achieved success by utilizing the pre-trained CLIP model [22] in downstream tasks, like text-based video retrieval [20], video captioning [28], and so on.

3 METHOD

In this section, we introduce our two stage multi-modal framework, which aims at tackling interaction prediction task in a unified framework. We first give an overview of the entire framework, then described sub-scene segmentation method. After that we introduce the use of multi-modal information (e.g., frames, audio and text). Finally, we present the training and inference of our method, meanwhile we also introduce the processing of query answering.

3.1 Pipeline

The pipeline of the proposed two stage multi-modal method is shown in Figure 2. For the first stage, we first try to divide long scene segment into several short sub-scenes, and obtain face trajectory by face tracking and recognition model [5, 8, 13] and person trajectory by skeleton-based person tracking method [33]. After that all face emotion feature $\mathcal{F}$ and skeleton-based person entity feature $\mathcal{P}$ can be extracted according to traces. Meanwhile, visual feature $\mathcal{V}$, text feature $\mathcal{T}$ and audio feature $\mathcal{A}$ of a scene segment are also extracted based on VIT [10], transformer [9] and MEF [21], respectively. We integrate these features as the input of next stage.

For the second stage, we first utilize zero-shot transfer model to predict interactions. Given the input video and a set of candidate interaction tags, we need to predict which sentiment tags are suitable for describing the video. We use the same pre-trained text encoder [9] to extract text features of provided interactions in a

scene. The similarity between integrated multi-modal features including $\mathcal{V}$, $\mathcal{T}$, $\mathcal{A}$, $\mathcal{F}$, $\mathcal{P}$ and features of predefined interaction are calculated.

Meanwhile, we also design a multiple object classification model [2] with BCEWithLogitsLoss to identify interaction categories based on the input of multi-modal features including $\mathcal{V}$, $\mathcal{T}$, $\mathcal{A}$. The final predicted result is fused with prediction results of two models.

3.2 Sub-Scene Segmentation

The purpose of sub-scene segmentation is to split long scene segment into multiple short clips using face recognition and person tracking, then features of face emotion and person trajectory in each clip can be extracted. Meanwhile, we assume that each clip contains only one interactive behaviour during testing.

The specific processing of sub-scene segmentation is as follows. We first sample frames of each scene by interval 10 frames and associates with corresponding timestamp information. Then, for each sampled frame, a face detector RetinaFace [8] together with a facial landmark localization PFLD [13] model are applied to detect face. In order to associate with the provided reference face, a face recognition model based on MobileFaceNet [5] is used. Finally, using Min-cost Network Flow Algorithm [27], the face trajectory of each entity with corresponding coordinates of face boxes are constructed.

In the scene segments, there are a serious masking of the face information. Consequentially, it is incomplete for sub-scene segmentation only using face trajectory. Considering the information of the human body can provide good complementarity for the masking of face, we simultaneously utilize a human body detection based on Faster R-CNN and a pose-based tracker based on HRNet to obtain person trajectory. Finally, by judging the overlap of coincidence between the person trajectory and face trajectory, we can obtain the complete trajectory for each entity in the scene segment.

3.3 Multi-modal Features

As shown in Figure 2, we use multi-modal features, including face emotion feature $\mathcal{F}$, skeleton-based person entity feature $\mathcal{P}$, visual feature $\mathcal{V}$, text feature $\mathcal{T}$ and audio feature $\mathcal{A}$ of a scene segment to predict interactions between different entities.

Face emotion feature Facing emotions, such as happiness, sadness, pain, etc., can reflect the interaction between characters and it is one of the important manifestations of interaction. Based on a pre-trained good facial emotion model AffectNet [25], facial emotion features are obtained by inferring all facial frames on the sub-scene segment.

Skeleton-based person entity feature For each person trajectory in sub-scene segment, we estimate 2D skeletons with a two-stage pose estimator (Faster-RCNN & HRNet) and directly provide the processed skeleton data as pickle files. Based on pre-trained action recognition model PoseC3D [12], we rely on a 2D heatmap stack instead of a graph sequence as the base representation of human skeletons to extract skeleton-based person entity feature.

Visual feature of scene segment We apply vision-language large pre-trained model (CLIP [22]) to extract spatial feature of each frame, and then fuse them in temporal level based on average pooling method.

Text feature of scene segment The dialogue text feature between the persons is one of the characteristics that can directly reflect the interactive relationship. Therefore, we first apply ASR to transform speech dialogue to text. Then, based on large-scale vision-language pre-training model, named Clip [22], we extract text feature using text transformer encoder [9].

Audio feature of scene segment According to Speechpy [29], we extract MFCC [18] and LMFE [16] features from the scene segmentation audio, and joint together to represent the audio features.

All the features are then concatenated together used as the input of multi-modal feature in zero-shot transfer model as f^m . However, because the linear classification layer in our multiple object classification model requires a fixed-scale input, we fuse visual feature, text feature and audio feature of scene as multi-modal feature with fixed dimension as f'^m .

3.4 Training And Inference

In this section, two models are proposed to predict interaction between different person entities.

For the first model, due to the limited samples and the absence of labels in the training set, we utilize zero shot transfer model to classify interactions as Clip [22]. Specially, the features of interaction vocabularies are generated using same way as the generation of features of ASR descriptions for a scene segment. During training, cosine similarity of multi-modal feature and the target interaction features are computed. We use a symmetric cross entropy loss over these similarity scores to train the model's parameters as follows.

$$\mathcal{L}_1 = -\frac{1}{B} \sum_{i=1}^B \log \frac{\sum_{j \in \mathcal{P}_i} \exp(s(f_i^m, f_j^l))}{\sum_{k \in \mathcal{N}_i} \exp(s(f_i^m, f_k^l))} \quad (1)$$

$$s(f_i, f_j) = \frac{f_i^T \cdot f_j}{\|f_i\| \|f_j\|} \quad (2)$$

In special, $\mathcal{L}_1$ represents symmetric cross entropy loss. B is the batch size during training. $\mathcal{N}_i$ denotes the set of negative interactions in the i scene segment, $\mathcal{P}_i$ denotes the set of positive interactions in the i scene segment, f_i^m is the multi-modal features of i scene segment, f_j^l is the text feature of interaction vocabularies in $\mathcal{P}_i$, and f_k^l is the text feature of interaction vocabularies in $\mathcal{N}_i$. The similarity function $s(f_i, f_j)$ is defined as the cosine similarity as equation 2.

During inference, the cosine similarity $s_i = s(f_i^m, f^l)$ of multi-modal feature and interaction feature is used as prediction score. f^l represents text feature of all interaction vocabularies.

For the second model, we use a linear classification layer with BCEWithLogitsLoss to classify interactions between different person entities. Specially, the input multi-modal feature of scene i is f_i^m , through linear classification layer, we get the prediction scores $o[i]$ of scene i . The training loss is as equation 3.

$$\mathcal{L}_2 = -\frac{1}{B} \sum_i (t[i] * \log(o[i])) + (1 - t[i]) * \log(1 - o[i]) \quad (3)$$

$t[i]$ represents the label of interactions in scene i .

During training, two models are trained respectively. During inference, we combine two prediction scores s_i and $o[i]$ with equal weights as final prediction scores.

Table 1: Comparison of different combinations of features with the proposed method on HLVU datasets.

Method	Feature	Spiritual	Honey	Nuclear	Sophie	Superhero	Finn	Shooters	Something	Time	Valkaama	Average	TestMovies
First	$\mathcal{V}/\mathcal{T}/\mathcal{A}/\mathcal{F}/\mathcal{P}$	52.5	26.5	67.8	56.2	36.8	31.2	62.3	77.5	53.2	74.0	53.8	-
Second	$\mathcal{V}$	22.2	16.9	12.5	29.9	22.7	20.2	39.5	35.3	42.3	64.0	30.5	-
	$\mathcal{T}$	31.0	13.9	19.1	30.9	22.7	15.4	31.6	37.5	40.8	61.2	30.4	-
	$\mathcal{A}$	25.3	12.8	11.5	24.5	20.1	15.4	22.6	24.8	33.6	55.5	27.1	-
	$\mathcal{V}/\mathcal{T}/\mathcal{A}$	53.5	27.8	66.7	55.7	36.4	30.8	63.2	77.8	52.9	74.9	53.4	-
Ensemble	$\mathcal{V}/\mathcal{T}/\mathcal{A}/\mathcal{F}/\mathcal{P}$	54.2	28.0	67.25	56.95	37.60	32.0	63.35	78.45	53.85	75.45	54.60	51.3

3.5 Query Answering

Based on the proposed two stage multi-modal method, we intend to answer three questions as Figure 1. Firstly, we divide a test scene segment into several shots according to section 3.2, consequence, each shot contains multiple person entities trajectory with corresponding timestamp. Secondly, we extract multi-modal features of each shot as Section 3.3 to predict the interactions. Different from training, we assume that there is only one interaction in each shot, and we get the maximum prediction score as final results during testing. Finally, for the first question "Finding the correct scene", we collect predicted interactions of all shots in a scene and then match most suitable scene with candidate answers based on maximum intersection principle. For the second question "Filling in the unknown character in the graph space", we traverse prediction results of all the shots for special scene and examine which person is satisfied. For the third question, "Finding next or previous interaction in special scene", as the same with second question, we traverse prediction results of all the shots for special scene and examine which interaction between the special person entities is satisfied under multiple options.

4 EXPERIMENTS

In this section, We first describe HLVU dataset and details of implementation. Then, we present more experimental results on HLVU datasets.

4.1 HLVU Dataset And Details Of Experiments

HLVU dataset contains 14 movies with total 16 hours. Each movie is divided into several scene segments, and each scene segment has manually annotated knowledge graph for that explains the relationship, interactions and sentiments between different entities in the scene. Meanwhile, the dataset also provides a set of reference images and entities for identifying each entity in the scene.

To evaluate the performance of our method, we follow [36] using metric $Recall@k$ as our evaluation methodology.

$$Recall@k = \frac{TP_k}{TP_k + FN_k} \quad (4)$$

k represents the correct predictions in k recalls.

In this paper, we conduct cross validation experiments on 14 movies. For example, we random select one movie as testing data and the rest 13 movies are training data. Following previous works [1], we report results on the 10 movies used as DVU 2021 train set. The face emotion feature and skeleton-based person entity feature are extracted by AffectNet [25] and PoseC3D [12], whose dimension

are 2048 and 512 respectively. In addition, visual feature, text feature and audio feature of a scene segment are extracted based on VIT [10], transformer [9] and MEF [21], respectively. The dimension of each feature is the same with 768. During fine-tuning of our network based on two methods, all batch size equal to 128, and the initial learning rates are set to 0.0001 respectively. Note that the dimension of f_i^m is fixed with 2304, but the scale of f_i^m is varied due to the uncertain number of entities in a scene. The number of classification label is 116 in multiple classification model.

4.2 Experimental Results

In order to validate the effectiveness of the proposed two stage multi-modal method, we construct different combinations of features to predict interactions for comparison.

As shown in Table 1, we first explore effect of each unimodal feature on the performance of interaction prediction based on the second multiple object classification model. The visual feature of scene is more helpful for predict interaction of scene compared with other unimodal feature, and other unimodal feature can also improve accuracy of predicting interactions as shown from second row to fourth row of Table 1. Combining visual feature with text and audio feature, it can be seen that the architecture of joint multi-modal feature such as $\mathcal{V}/\mathcal{T}/\mathcal{A}$ has benefit in predicting interactions. The performance is improved by 22.9% compared with only using visual feature of scene.

Besides that we fuse the predicted results of first zero-shot transfer model with the results of second multiple object classification model as Ensemble model shown in Table 1. In the ensemble architecture, we achieve the best results on HLVU dataset, which proves the effectiveness of multi-modal features and ensemble model for our method. We further train our framework on all training data of HLVU dataset, the final results on test movies are shown in Table 1. The proposed model obtains 51.3% accuracy on the test movies, which also achieves comparable performance with our validation results.

5 CONCLUSION

In this paper, we propose a two stage multi-modal method to answer the entity interactive questions in a scene. This framework can benefit from the large scale pre-trained zero-shot transfer models and has advantage of multiple classification model. Besides that, multi-modal features extractor aggregated face emotion, person skeleton, video, text and audio modal informations to together evaluate interaction categories. Multi-modal features are unified in the same semantic space. The performances on the HLVU dataset show the effectiveness of our method.

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